

TABLE I | Two service capacities across ten CIM technologies

INT8 logical payloads · Selected native configurations · Sustained complete-matrix loading

Technology / native K × N / update	Optimistic / fast			Typical			Pessimistic / slow		
	ρ [MB/s]	τ [MB/s]	RI* [-]	ρ [MB/s]	τ [MB/s]	RI* [-]	ρ [MB/s]	τ [MB/s]	RI* [-]
SRAM ACIM 128×128 · 16 B groups	83	7,300	0.011	40	3,200	0.012	17	1,600	0.010
SRAM DCIM 128×16 · 16 B groups	460	7,300	0.063	390	3,200	0.12	190	1,600	0.12
2D NOR 128×128 · sector/page overwrite	34	0.080	420	24	0.080	300	18	0.0091	2,000
3D NAND 4608×480 · block/page replace	17	1.2	14	14	1.2	12	11	0.56	20
RRAM 128×64 · 8 KiB epoch; 16 B groups	48	7.1	6.7	25	4.3	5.8	12	1.7	7.0
MRAM 256×32 · 8 B groups	750	130	5.6	320	76	4.1	160	53	3.0
PCM 256×128 · 32 B row stripes	5.2	8.0	0.65	2.5	6.9	0.36	1.0	4.9	0.21
HfO₂ FeRAM 128×128 · 16 B groups	69	310	0.22	27	110	0.25	13	53	0.24
Gain-cell eDRAM 64×64 · 16 B groups	15	200	0.076	9.0	170	0.054	5.4	120	0.047
3D FeFET 128×128 · 16 B groups	100	81	1.3	47	60	0.78	23	41	0.57

ρ : complete input-vector evaluation capacity. τ : full resident-matrix loading capacity. RI* = ρ/τ .

Fast / typical / slow are paired, sustainable conditions in one configuration; resource expansions and refresh stress points are excluded.

K × N counts input and output elements. Group labels describe local update shape; reported τ includes all groups and required full-load overhead.

Gain-cell rates include refresh. NOR / NAND include erase. Native configurations differ in resources and size; this is not an equal-area or equal-work ranking.

NAND rates are conditional: offset-coded quantization can reverse weak or canceling outputs; small-signal accuracy is not guaranteed.